31st March, 2026

Dear Professor Buchanan and Dr Matzke,

Please find attached an original research submission entitled *"CzeLex: A megastudy dataset of concreteness, imageability, age of acquisition, valence and gender ratings for 9,601 Czech words"*, co-authored by James Brand, Filip Smolík, Jan Chromý, Markéta Cehánková and Mikuláš Preininger.

This paper presents the first large-scale set of semantic norms that focus on the multidimensional representation of meaning, specifically the extent to which words are associated to scales of Age of Acquisition, Concreteness, Gender, Imageability, and Valence. Along with the manuscript, we make available raw and summary versions of our data, as well as all analysis scripts to make the resources more accessible to a wide range of research applications. We believe this manuscript and the associated materials are a novel and valuable contribution to the behavioural and cognitive sciences, and that Behaviour Research Methods is the ideal outlet for the work.

In the manuscript, we provide detailed information on the methods, pre-processing and filtering of the data, along with a range of summary variables for 9,601 Czech words. The norms are then analysed and validated through reliability measures, correlations between the variables, as well as assessing the extent to which they predict measures of lexical processing. An additional novel contribution of the paper is the analysis of different aggregation methods, exploring whether there are differences across traditional summary statistics (such as means and SDs) and latent estimates using ordinal mixed-effects modelling.

All data, analyses and scripts will be made publicly available following publication, with a link available for reviewers at: https://osf.io/tqxfm

All research involving human participants has been carried out with appropriate ethical considerations and clearance from relevant ethical review panels. This work represents primary scientific research and is not under consideration for publication elsewhere. Its submission is approved by all authors.

We thank you for considering this paper and we look forward to hearing from you in due course.

Yours sincerely,

James Brand, Filip Smolík, Jan Chromý, Markéta Cehánková & Mikuláš Preininger

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